Pairing-Symmetry Selection Rules for the Cooper-Pair Mass Anomaly from A_5 Microsector Representation Theory

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The Cooper-pair mass in niobium, measured by Tate *et al.* (1989) via the London moment, exceeds twice the free-electron mass by $\delta = 92 \pm 21\,\text{ppm}$—an anomaly unexplained for 36 years. Working within the A_5 microsector of Density Field Dynamics (DFD), we establish three results at decreasing levels of rigor. *First* (exact within the pair-space construction): two pairing-symmetry selection rules. (a) An angular-cancellation rule: the quintet exchange channel in $S^2(V^*) = \mathbf{1} \oplus \mathbf{5}$ couples maximally to s-wave condensates but vanishes for d-wave condensates, whose sign-changing gap produces destructive interference. (b) An orthogonality rule: spin-triplet pairs live in $\Lambda^2(V^*) = \mathbf{3}$, which is orthogonal to the quintet by representation theory alone, independent of gap structure. *Second* (mechanism conjecture): the physical Cooper pair $|00\rangle$ carries quintet weight 2/3, accessing an exchange channel unavailable to uncorrelated single electrons, at the natural scale $\mathcal{O}(\alpha^2)$. *Third* (numerical conjecture): a motivated coefficient $\delta = \sqrt{3}\,\alpha^2 = 92.23\,\text{ppm}$, with $\sqrt{3}$ from the three-generation structure of $\mathbb{C}P^2 \times S^3$. Unlike the BCS-exchange correction of Lipavský (2016), which is material-dependent, the framework predicts universality for conventional s-wave superconductors—a distinction testable with existing technology.

INTRODUCTION

A rotating superconductor generates a magnetic field $\boldsymbol{B} = -(2m^*/e^*)\,\boldsymbol{\omega}$ (the London moment), where m^* is the effective Cooper-pair mass and $e^* = 2e$ [1, 2]. Tate *et al.* at Stanford, using a rotating niobium ring with SQUID flux detection, measured [3, 4]

$$\left(\frac{m^*}{2m_e}\right)_{\text{Nb}} = 1.000\,084\,(21)\,. \tag{1}$$

Standard relativistic and lattice-potential corrections predict $m^*/2m_e = 0.999\,992$ [5, 6], giving a residual anomaly

$$\delta_{\text{Tate}} = 92 \pm 21\ \text{ppm}\,, \tag{2}$$

a 4.4σ discrepancy in the wrong direction (mass increase rather than decrease).

Three theoretical responses exist. Tajmar and de Matos [7] attributed the anomaly to an enhanced gravitomagnetic London moment, but used δ_{Tate} as input; precision tests subsequently ruled out gravitomagnetic coupling at the required magnitude [8, 9]. De Matos [10] invoked a graviton mass inside the condensate, also using δ as input. Most substantively, Lipavský [11] showed that Pauli exchange processes—partner-swapping within the BCS condensate—produce a positive relativistic mass correction, reversing the sign predicted by earlier BEC-type calculations. His result matches Tate's magnitude for niobium, but the correction enters through the crystal work function ($W \approx 4\,\text{eV}$ for Nb) and the pairing strength Δ/E_F, making it material-dependent.

No existing theory provides all three of: (a) pairing-symmetry selection rules with distinct predictions for s-wave, d-wave, and p-wave materials, (b) material independence for a defined class of superconductors, and (c) a numerical prediction for the anomaly magnitude.

In this Letter we present three results, with their epistemic status explicitly labeled following the claim taxonomy of Ref. [12]: selection rules exact within the A_5 pair-space construction, a mechanism conjecture identifying the exchange channel, and a numerical conjecture for the coefficient.

FRAMEWORK

Density Field Dynamics (DFD) [12] replaces spacetime curvature with a scalar refractive field ψ on flat $\mathbb{R}^{3,1}$, with optical metric $d\tilde{s}^2 = -c^2 e^{-2\psi}\,dt^2 + d\boldsymbol{x}^2$ and refractive index $n = e^{\psi}$. The internal manifold $X = \mathbb{C}P^2 \times S^3$ yields the Standard Model gauge group, three fermion generations from $c_2(\mathbb{C}P^2) = 3$, and the fine-structure constant $\alpha^{-1} = 137.036$ from Chern–Simons quantization on S^3 with $k_{\max} = |A_5| = 60$ [12].

The A_5 microsector assigns each fermion generation to the fundamental three-dimensional representation V^* of A_5 (the icosahedral rotation group). The finite Yukawa operator [12] acts on the group algebra $\mathbb{C}[A_5]$ via Cayley-graph matrix elements and conjugacy-class projectors, producing the fermion mass hierarchy with 1.9% mean accuracy across nine charged fermions [12].

The gravitational clock coupling for a single electron arises at one loop [12]:

$$k_\alpha = \frac{\alpha^2}{2\pi} \approx 8.48 \times 10^{-6}\,, \tag{3}$$

where one factor of α is the ψ–EM vacuum vertex strength and $\alpha/(2\pi)$ is the Schwinger anomalous magnetic moment [13].

RESULT I: PAIRING-SYMMETRY SELECTION RULES

(Exact within the A_5 pair-space construction)

A Cooper pair is a two-electron bound state. In the A_5 microsector, each electron's generation quantum number lives in V^* ($\dim V^* = 3$). The pair's internal microsector state lives in the tensor product $V^* \otimes V^*$, which decomposes as

$$V^* \otimes V^* = S^2(V^*) \oplus \Lambda^2(V^*)\,, \tag{4}$$

with

$$S^2(V^*) = \mathbf{1} \oplus \mathbf{5}\,, \tag{5}$$

$$\Lambda^2(V^*) = \mathbf{3}\,. \tag{6}$$

The quintet $\mathbf{5}$ in $S^2(V^*)$ is the exchange channel relevant to the mass anomaly. Which component of $V^* \otimes V^*$ is physically realized depends on the overall exchange symmetry of the full pair wavefunction $\Psi_{\text{pair}} = \psi_{\text{orb}} \otimes \chi_{\text{spin}} \otimes \phi_{\text{internal}}$, which must be antisymmetric under particle exchange.

Spin-singlet pairs (s-wave and d-wave)

In conventional BCS superconductors and in cuprate d-wave superconductors, pairing is spin-singlet [14]: the spin part χ_{spin} is antisymmetric. Both s-wave ($\ell = 0$) and d-wave ($\ell = 2$) orbital wavefunctions are symmetric under exchange. Overall antisymmetry then requires the internal microsector part ϕ_{internal} to be *symmetric*, placing both cases in $S^2(V^*)$.

The distinction between s-wave and d-wave therefore arises not from the A_5 decomposition, but from how the gap symmetry enters the coupling to the exchange channel.

For an s-wave condensate, the gap $\Delta_{\boldsymbol{k}} = \Delta\, e^{i\theta}$ is isotropic. The anomalous propagator $F(\boldsymbol{k}) = \langle c_{\boldsymbol{k}\uparrow} c_{-\boldsymbol{k}\downarrow}\rangle$ locks all angular channels to the same macroscopic phase θ. When the exchange operator acts on the pair, every angular direction contributes coherently, and the coupling to the quintet is maximal.

For a d-wave condensate, $\Delta_{\boldsymbol{k}} = \Delta_0 \cos 2\phi_{\boldsymbol{k}}$, which changes sign at four nodal directions. The angular integral governing the coherent coupling becomes

$$\int_0^{2\pi} \frac{d\phi}{2\pi} \cos 2\phi = 0\,. \tag{7}$$

The sign-changing gap produces destructive interference that cancels the exchange coupling.

Selection Rule 1 (angular cancellation).—Within the A_5 pair-space construction, the quintet exchange channel of $S^2(V^*)$ produces a nonzero ψ-coupling for s-wave condensates and a vanishing coupling for d-wave condensates. This rule is exact given the A_5 decomposition and the condensed-matter input of the gap symmetry; it requires both ingredients.

Spin-triplet pairs (p-wave)

For spin-triplet pairing (p-wave, $\ell = 1$), the situation is fundamentally different. The orbital wavefunction is antisymmetric, the spin part is symmetric (triplet), so overall antisymmetry requires the internal microsector part to be *antisymmetric*: the pair lives in $\Lambda^2(V^*) = \mathbf{3}$.

Since $\Lambda^2(V^*)$ and $S^2(V^*)$ are orthogonal subspaces of $V^* \otimes V^*$, the triplet pair has exactly zero projection onto the quintet $\mathbf{5}$:

$$\langle \mathbf{5} \,|\, \Pi_{\Lambda^2} \,|\, \mathbf{5}\rangle = 0\,. \tag{8}$$

Selection Rule 2 (representation orthogonality).—Spin-triplet pairs live in $\Lambda^2(V^*)$, which is orthogonal to the quintet exchange channel by A_5 representation theory alone. The anomalous mass correction vanishes identically for spin-triplet superconductors, independent of gap symmetry, angular structure, or any condensed-matter input beyond the spin state.

This is a purely representation-theoretic result, requiring no condensed-matter input beyond the identification of the spin state.

Experimental status

Selection Rule 1 is consistent with prior null reports from Tajmar [15] for YBCO and BSCCO at 77 K, and from Chiao [16] for YBCO—both d-wave, spin-singlet materials. These experiments differ in method and precision from Tate's London-moment measurement, so the comparison is qualitative rather than a direct replication of the Tate observable. Selection Rule 2 predicts a null for any candidate spin-triplet superconductor; this has not yet been tested. Neither selection rule is predicted by Lipavský [11], whose mechanism has no representation-theoretic content and does not address pairing symmetry.

RESULT II: PAIR-ONLY EXCHANGE CHANNEL

(Mechanism conjecture)

In a conventional BCS superconductor, both electrons in a Cooper pair occupy generation 1. The pair's microsector state is $|00\rangle \in S^2(V^*)$, which decomposes as

$$|00\rangle = \tfrac{1}{\sqrt{3}}\,|\mathbf{1}\rangle + \sqrt{\tfrac{2}{3}}\,|\mathbf{5}\rangle\,. \tag{9}$$

The democratic singlet $|\mathbf{1}\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |11\rangle + |22\rangle)$ is generation-blind and carries no exchange residual beyond tree level. A single uncorrelated electron occupies only

one generation direction and has zero projection onto the quintet (which requires a two-particle state to exist).

The quintet component, with weight 2/3, is the pair-only exchange channel: it is accessible to Cooper pairs but not to single electrons. Any operator coupling through the **5** of $S^2(V^*)$ produces a signal for s-wave pairs that is absent for isolated electrons.

The natural scale of this coupling is set by the ψ–EM vertex structure: one factor of α from the vertex, one from the loop, giving $\mathcal{O}(\alpha^2) \sim 5.3 \times 10^{-5}$, i.e., of order 50–100 ppm. Tate's anomaly at 92±21 ppm falls squarely in this range.

We note what this does *not* establish. The map from the finite Yukawa operator (acting on $\mathbb{C}[A_5]$) to the London-moment observable (measured by a SQUID on a rotating ring) requires an auxiliary bridge connecting the microsector pair state to the constitutive mass m^* in $\boldsymbol{B} = -(2m^*/e^*)\boldsymbol{\omega}$. This bridge is physically motivated but not yet theorem-grade.

RESULT III: COEFFICIENT

(Numerical conjecture)

We conjecture that the anomalous mass correction for s-wave Cooper pairs is

$$\boxed{\delta = \sqrt{3}\,\alpha^2 = 92.23 \text{ ppm}} \tag{10}$$

with $\sqrt{3} = \sqrt{N_{\text{gen}}}$ and $\alpha^{-1} = 137.035\,999\,084$.

The coefficient arises from three factors:

(i) *ψ–EM vertex*: α.

(ii) *Coherence-enhanced loop*: α (with the $1/(2\pi)$ angular-averaging suppression of the free-electron Schwinger diagram lifted by the macroscopic phase coherence of the s-wave condensate, which locks all angular channels to the same macroscopic phase via the anomalous propagator).

(iii) *Generation factor*: $\sqrt{N_{\text{gen}}} = \sqrt{3}$ (all three generation channels contribute, with topologically fixed phases adding incoherently in amplitude).

The match to Eq. (2) is 0.01σ.

We assign this result the status of *numerical conjecture* for three reasons. First, the coherence-lifting argument (replacing $1/(2\pi)$ with 1) is physically motivated by the off-diagonal long-range order of the BCS ground state but has not been derived from a controlled diagrammatic calculation within the DFD framework. Second, the $\sqrt{N_{\text{gen}}}$ factor assumes incoherent amplitude addition of generation channels, which is plausible but unproven. Third, exploratory numerical scans of the full 60-dimensional $\mathbb{C}[A_5]$ group algebra yield coefficients in the range 1.75–1.78 (close to $\sqrt{3} \approx 1.732$) for specific operator assignments, but these depend on the choice of eigenbasis within the quintet subspace and are not yet basis-independent.

The coefficient involves no continuous free parameters once the A_5 auxiliary closure is adopted. The discrete structural choices (generation assignment, operator path on the Cayley graph) are constrained by the microsector framework but not uniquely fixed by the core DFD field equations alone.

EXPERIMENTAL DISCRIMINATION

The sharpest test distinguishing this framework from Lipavský [11] is universality.

Lipavský's BCS-exchange correction enters through the work function and pairing strength, both of which vary across materials (Table I). The gap ratio Δ/E_F spans nearly a factor of 20 from Al to Nb. If the anomaly tracks Δ/E_F, Lipavský is confirmed and DFD is falsified. If the anomaly is the same (≈ 92 ppm) for all six materials, DFD survives.

The decisive experiment is a multi-material London-moment measurement at $\lesssim 20$ ppm precision. Tate's 1989 measurement of niobium [3, 4] remains the only precision determination; it has never been replicated or extended to other materials. Modern SQUID magnetometry should permit sub-ppm accuracy [17], though no such measurement has yet been realized.

Additional predictions (Table II):

(i) *d-wave null*: $\delta = 0$ for all d-wave spin-singlet superconductors, consistent with prior null reports [15, 16].

(ii) *p-wave null*: $\delta = 0$ for spin-triplet superconductors by representation orthogonality. Untested.

(iii) *Clock test*: An optical lattice clock at radius r inside a rotating Nb ring sees $\delta f/f \sim 2\delta\,\omega\,r\,\alpha/(2c) \sim 8 \times 10^{-15}$ at $\omega = 100$ rad/s, $r = 3.6$ cm—above current Sr clock sensitivity [19].

(iv) *Equivalence principle*: Ross *et al.* [20] bound the Cooper-pair Eötvös parameter to $\eta_{\text{CP}} \leq 9.2 \times 10^{-4}$. The DFD prediction $\delta \approx 9.2 \times 10^{-5}$ is within reach of a one-order-of-magnitude improvement.

TABLE I. Superconducting gap Δ and Fermi energy E_F for six type-I superconductors. The framework predicts the same δ for all; Lipavský's mechanism predicts δ varying with Δ/E_F.

Material	Δ (meV)	E_F (eV)	Δ/E_F
Nb	1.55	5.32	2.9×10^{-4}
Pb	1.35	9.47	1.4×10^{-4}
Sn	0.59	10.2	0.6×10^{-4}
In	0.54	8.63	0.6×10^{-4}
Al	0.17	11.7	0.15×10^{-4}
Hg	0.82	7.13	1.2×10^{-4}

TABLE II. Selection rule predictions by pairing symmetry.

Pairing	Example	Mechanism	δ
s-wave singlet	Nb, Pb, Al	Quintet channel	$\mathcal{O}(\alpha^2)$
d-wave singlet	YBCO, BSCCO	Angular cancel.	0
p-wave triplet	(candidates)	Λ^2 orthog.	0

DISCUSSION

Table III summarizes the comparison. The unique contribution of this work is the pair of selection rules: a sharp, parameter-free distinction between s-wave, d-wave, and p-wave materials. These results survive independently of the numerical coefficient.

An open question is the relationship between Lipavský's "Pauli exchange" and our "quintet exchange channel." Both identify exchange processes within the condensate as the physical origin. They may describe the same physics in different formalisms. If so, the DFD contribution is to explain *why* the exchange correction takes its magnitude (group theory) and *why* certain pairing symmetries give zero (angular cancellation and representation orthogonality). Computing Lipavský's correction explicitly for multiple materials would resolve this.

We stress that Eq. (1) is a single measurement from 1989, at 21 ppm accuracy, that has never been replicated. The 0.01σ agreement with $\sqrt{3}\,\alpha^2$ is striking but constitutes motivation for replication, not confirmation.

The principal theoretical gap is the auxiliary bridge connecting the A_5 pair-space operator to the London-moment observable: the map from the finite Yukawa matrix element $\langle\chi_R|Y_{\text{finite}}|\chi_L\rangle$ acting on $\mathbb{C}[A_5]$ to the constitutive mass m^* in the London equation. Closing this bridge would elevate the mechanism conjecture to a theorem.

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TABLE III. Comparison of theoretical approaches. "Input" means δ_{Tate} was used to fix a parameter.

Theory	Prediction (ppm)	Material indep.?	Free params	Selection rules?
Tajmar [7]	92 (input)	No	1	No
Lipavský [11]	∼92	No	≥1	No
de Matos [10]	92 (input)	No	1	No
This work				
Sel. rules	0 (d/p)	Yes	0	**Yes**
Coefficient	92.23*	Yes	0†	—

*Numerical conjecture. †No continuous free parameters once auxiliary closure is adopted; see text.

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